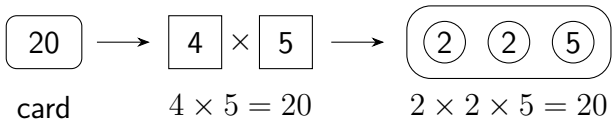
A number card can split into two whole numbers greater than 1 that multiply to its value. These numbers are its factors. Keep splitting until every card is prime: a number greater than 1 with no factors except 1 and itself. The finished cards make a prime bag. Multiply its tokens to find its value.



Token order does not matter. Equal tokens are interchangeable.

Problem 1: For each card, find two different splitting routes. Can different routes finish with different prime bags?

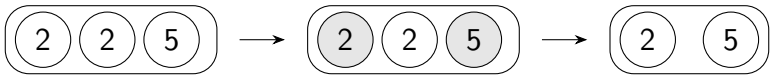
12

18

24

36

Choose some tokens from a bag, using each token at most once. The chosen tokens form a bag. You may choose all of them or none. An empty bag has value 1. A divisor of a whole number divides it with no remainder.



value 20 choose shaded tokens value 10

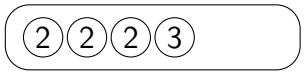
$2 \times 5 = 10$ $20 \div 10 = 2$

Problem 2: For each bag, find every different value you can make by choosing tokens. Does your collection contain every divisor of the full bag's value?

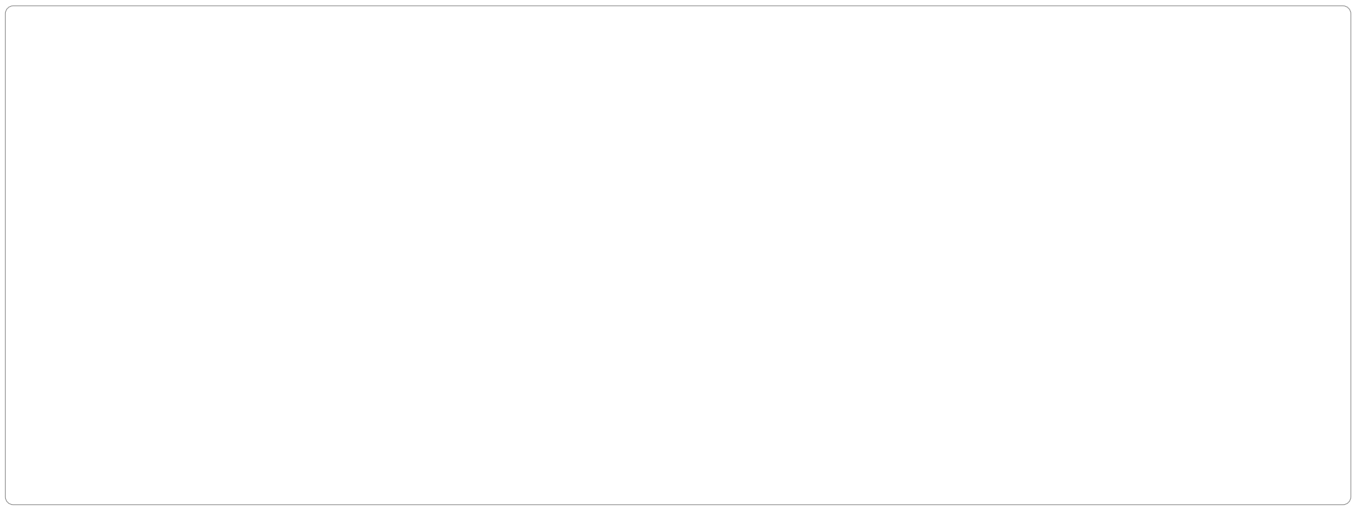
(2)(2)(3) value 12

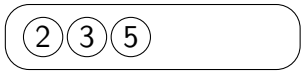
(2)(3)(3) value 18

Problem 3: These bags have different numbers of tokens. Which can make more different values by choosing tokens? Find the complete collection for each bag.

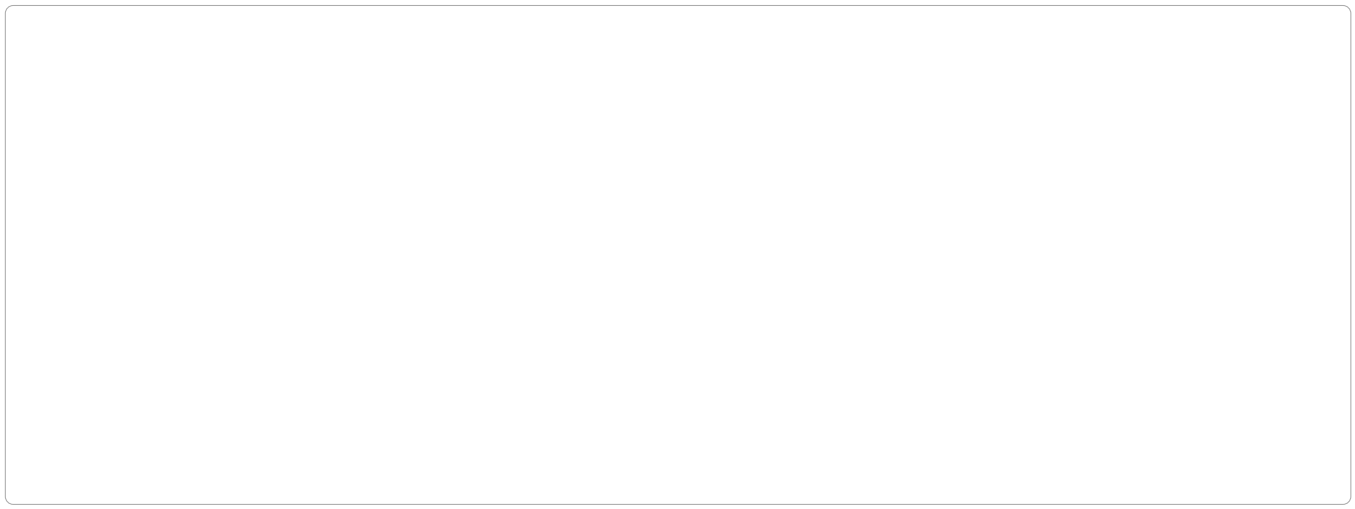


value 24





value 30



Problem 4: Use prime tokens with any of the labels 2, 3 and 5. Build a bag that makes exactly 4 different values by choosing tokens, another that makes exactly 6, and another that makes exactly 8. How small can you make the full bag's value in each case?

4 different values

6 different values

8 different values